

1. dataset_loaded.png

```
[9]: # 1. Change the extension to .csv
file = wslib.load_data("predictive_maintenance.csv")

# 2. Change read_excel to read_csv
df = pd.read_csv(file)

# 3. View the data
df.head()
```

	UDI	Product ID	Type	Air temperature [K]	Process temperature [K]	Rotational speed [rpm]	Torque [Nm]	Tool wear [min]	Target	Failure Type
0	1	M14860	M	298.1	308.6	1551	42.8	0	0	No Failure
1	2	L47181	L	298.2	308.7	1408	46.3	3	0	No Failure
2	3	L47182	L	298.1	308.5	1498	49.4	5	0	No Failure
3	4	L47183	L	298.2	308.6	1433	39.5	7	0	No Failure
4	5	L47184	L	298.2	308.7	1408	40.0	9	0	No Failure

2. data_info.png

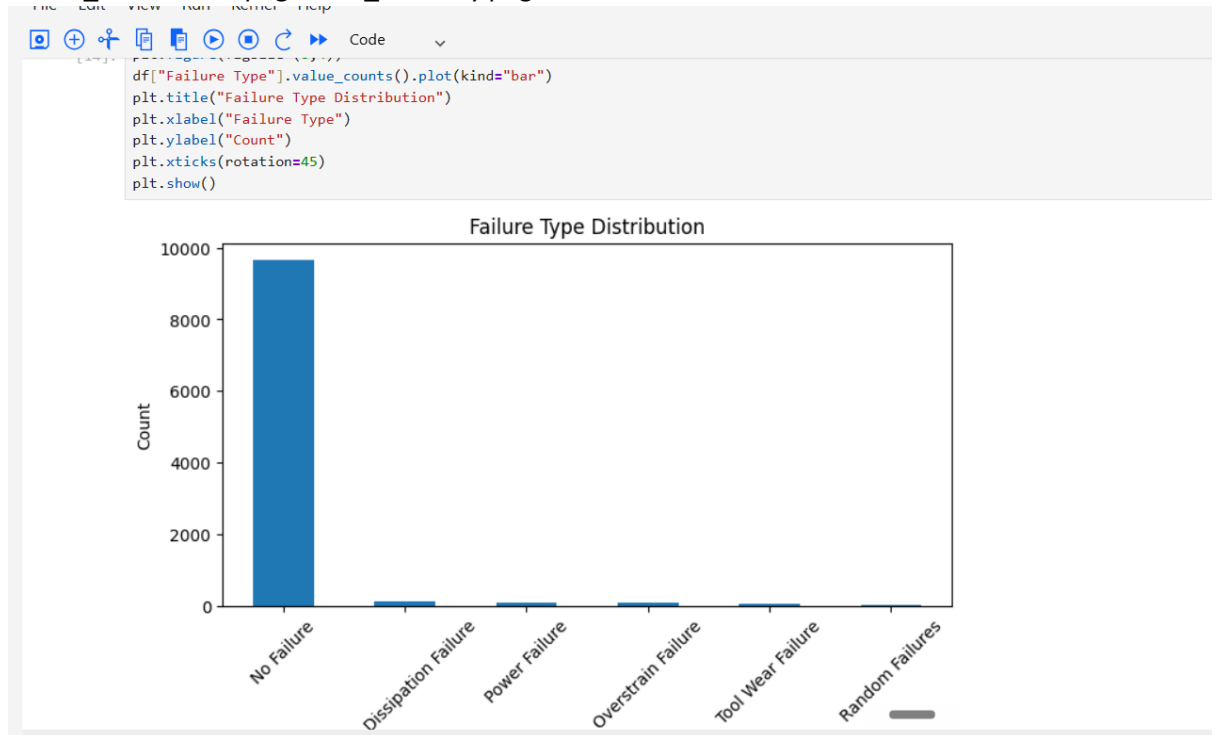
```
[10]: df.shape

[10]: (10000, 10)

[11]: df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 10 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   UDI                                    10000 non-null  int64
1   Product ID                            10000 non-null  object
2   Type                                  10000 non-null  object
3   Air temperature [K]                   10000 non-null  float64
4   Process temperature [K]                10000 non-null  float64
5   Rotational speed [rpm]                 10000 non-null  int64
6   Torque [Nm]                           10000 non-null  float64
7   Tool wear [min]                       10000 non-null  int64
8   Target                                10000 non-null  int64
9   Failure Type                           10000 non-null  object
dtypes: float64(3), int64(4), object(3)
memory usage: 781.4+ KB
```

3. failure_distribution.pngmodel_accuracy.png



4. model_accuracy.png

```
Model training completed successfully

[21]: y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)

print("Model Accuracy:", accuracy)

Model Accuracy: 0.9815
```

5. classification_report.png

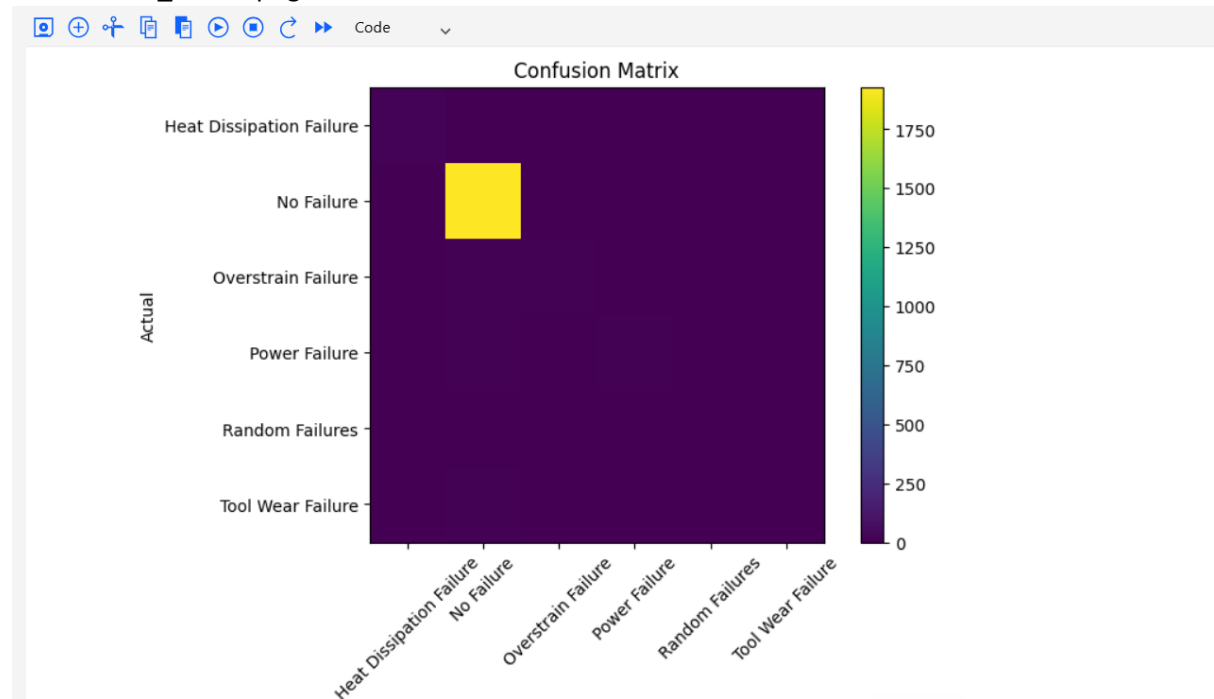
```
print("Model Accuracy:", accuracy)

Model Accuracy: 0.9815

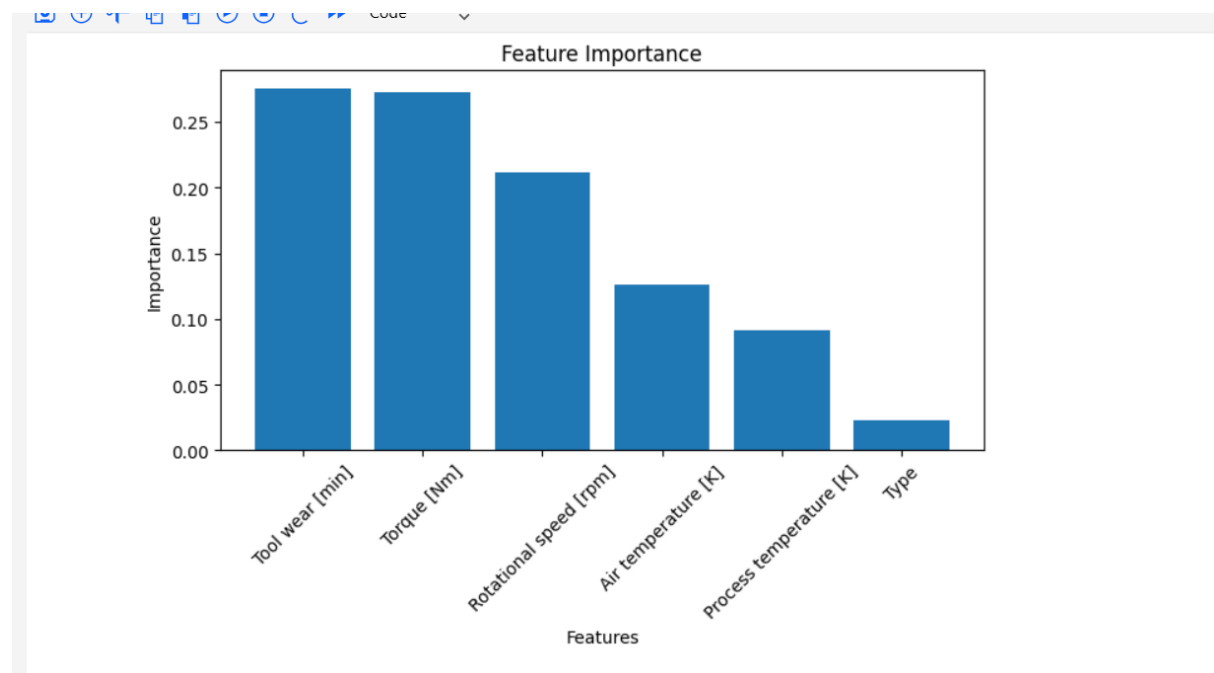
[22]: print(classification_report(y_test, y_pred, target_names=le_failure.classes_))
```

	precision	recall	f1-score	support
Heat Dissipation Failure	1.00	0.82	0.90	22
No Failure	0.98	1.00	0.99	1930
Overstrain Failure	1.00	0.50	0.67	16
Power Failure	0.89	0.42	0.57	19
Random Failures	0.00	0.00	0.00	4
Tool Wear Failure	0.00	0.00	0.00	9
accuracy			0.98	2000
macro avg	0.65	0.46	0.52	2000
weighted avg	0.97	0.98	0.98	2000

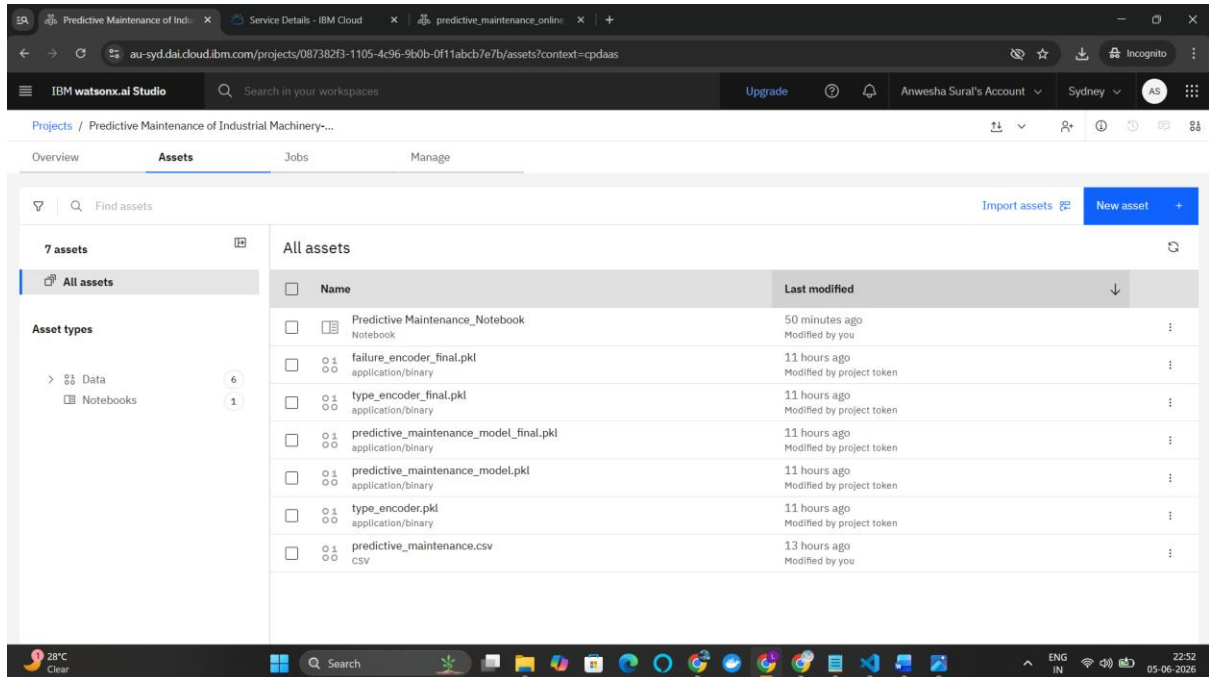
6. confusion_matrix.png



7. feature_importance.png



8. ibm_assets.png



9. model_stored.png

```
Model retrained successfully
Accuracy: 0.9815

In [21]: software_spec_uid = client.software_specifications.get_id_by_name("runtime-25.1-py3.12")

model_props = {
    client.repository.ModelMetaNames.NAME: "predictive_maintenance_rf_model",
    client.repository.ModelMetaNames.TYPE: "scikit-learn_1.6",
    client.repository.ModelMetaNames.SOFTWARE_SPEC_UID: software_spec_uid
}

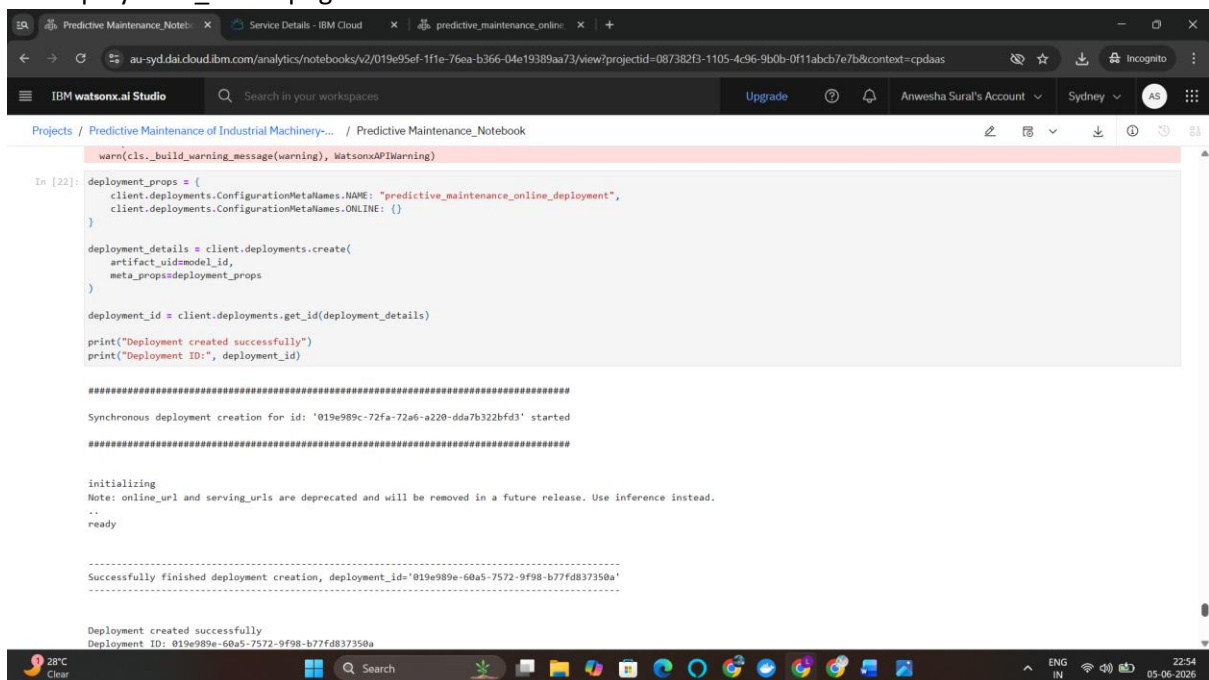
stored_model_details = client.repository.store_model(
    model=model,
    meta_props=model_props,
    training_data=train,
    training_target=train_target
)

model_id = client.repository.get_model_id(stored_model_details)

print("Model stored successfully")
print("Model ID:", model_id)

Model stored successfully
Model ID: 019e989c-72fa-72a6-a220-dda7b322bfd3
```

10. deployment_status.png



```
Deployment created successfully
Deployment ID: 019e989e-60a5-7572-9f98-b77fd837350a

In [23]: deployment_details = client.deployments.get_details(deployment_id)

Out[23]: {'entity': {'asset': {'id': '019e989c-72fa-72a6-a220-dda7b322bfd3'},
  'chat_enabled': False,
  'custom': {},
  'deployed_asset_type': 'model',
  'hardware_spec': {'name': 'S', 'num_nodes': 1},
  'online': {},
  'status': {'inference': [{'url': 'https://au-syd.ml.cloud.ibm.com/ml/v4/deployments/019e989e-60a5-7572-9f98-b77fd837350a/predictions'}],
  'online_url': {'url': 'https://au-syd.ml.cloud.ibm.com/ml/v4/deployments/019e989e-60a5-7572-9f98-b77fd837350a/predictions'},
  'serving_urls': ['https://au-syd.ml.cloud.ibm.com/ml/v4/deployments/019e989e-60a5-7572-9f98-b77fd837350a/predictions'],
  'state': 'ready'}],
  'metadata': {'created_at': '2026-06-05T16:29:36.326Z',
  'id': '019e989e-60a5-7572-9f98-b77fd837350a',
  'modified_at': '2026-06-05T16:29:36.326Z',
  'name': 'predictive_maintenance_online_deployment',
  'owner': 'IBMid-691001MHU2',
  'space_id': 'def5c3a9-1d49-4668-a45f-50db938b91f5',
  'system': {'warnings': [{'id': 'Deprecated',
  'message': 'online_url and serving_urls are deprecated and will be removed in a future release. Use inference instead.'}]}}
```

```
In [24]: sample_payload = {
  client.deployments.ScoringMetalNames.INPUT_DATA: [
    {
      "fields": list(X.columns),
      "values": [
        [0, 298.1, 308.6, 1551, 42.8, 0]
      ]
    }
  ]
}
```

11. api_reference.png

Deployment spaces /

predictive-maintenance-space

Overview Assets **Deployments** Jobs Manage

Name	Type	Status	Asset	Asset type	Tags	Last modified	
 predictive_maintenance_online_deployment	Online	 Deployed	predictive_maintenance_rf_model	Model		55 minutes ago Anwesha Sural (You)	

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12. prediction_result.png

